

②

$$y = \underset{\textcircled{1}}{AB'C} + \underset{\textcircled{2}}{A'BC} + \underset{\textcircled{3}}{ABC'} + \underset{\textcircled{4}}{A'B'C} + \underset{\textcircled{5}}{AB'C'}$$

combine ①, ④ :

$$AB'C + A'B'C = B'C$$

combine ③, ⑤ :

$$ABC' + AB'C' = AC' \quad \textcircled{2} \rightarrow \text{remain}$$

$$\Rightarrow y = A'BC + B'C + AC'$$

$$y = (A'B + B')C + AC'$$

$$\Downarrow$$
$$y = (A' + B')C + AC'$$

$$y = A'C + AC' + B'C$$

$$y = A \text{ XOR } C + B'C$$

num of gates = 1 XOR, 1 OR, 1 AND, 1 NOT
4 gates.